4.1 Introduction

The essential elements for the existence of human beings are air, water, food, shelter, clothing, etc. Out of the above, water is the second element. Water is required for the satisfactory physiological functioning of living organisms, as a circulatory fluid to maintain temperature, to carry nourishing food and to remove the waste products from the body.

It is necessary that, water must be good and it should be free from impurities, toxic chemical compounds and pathogens. Therefore, in order to ensure the availability of sufficient quantity of good quality of water, it is essential to plan and build suitable water supply schemes.

4.1.1 Objectives of Public Water Supply Schemes

Following are the objectives of public water supply scheme:

1. To supply safe, wholesome water to the people so as to keep the diseases away and thereby promoting better health.
2. To maintain better sanitation and beautification of surroundings.
3. To ensure safety against fire, by supplying sufficient quantity of water.
4. To provide industrialisation and modernisation of the society and thereby ensuring better living standards.
5. To promote wealth and welfare for the entire community.

4.1.2 Planning of Water Supply Scheme

The water supply scheme for a city is prepared by the combination of field observation and office work. Following are the points of importance in any water supply scheme.

1. Population

From the available data of census of the last two decades, the future population may be predicted by any suitable method. Based on the future expansion, the water requirement for the scheme is decided.

2. Per Capita water demand

The rate of consumption of water per capita should be decided by taking all the possible uses of water as domestic, industrial, public, trade, fire demand, etc. The rate when multiplied by the population gives total quantity of water required for water supply schemes.

3. Sources of Water supply

The success of a water supply scheme entirely depends upon a good source of supply of water. The source should provide adequate and good quality of water throughout the year.

4. Quality of water

The quality of available water decides the line of treatment of water. The cost of treatment depends on its quality. If the water is pure, lesser will be the cost of treatment

5. Financial aspects

Based on the fund availability, the scheme should be adjusted and it should be as economical as possible.

6. Topography of the area

The topographical map of the area should be prepared. Low lying area, plain area, ridges, high density of population area should be marked on the map and analysed to ensure a simple and cheap water supply scheme.

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